

Business Problem

In recent years, City Hotel and Resort Hotel have seen high cancellation rates. Each hotel is now dealing with a number of issues as a result, including fewer revenues and less than ideal hotel room use. Consequently, lowering cancellation rates is both hotels' primary goal in order to increase their efficiency in generating revenue, and for us to offer thorough business advice to address this problem. The analysis of hotel booking cancellations as well as other factors that have no bearing on their business and yearly revenue generation are the main topics of this report.

Assumptions

1. No unusual occurrences between 2015 and 2017 will have a substantial impact on the data used.
2. The information is still current and can be used to analyze a hotel's possible plans in an efficient manner.
3. There are no unanticipated negatives to the hotel employing any advised technique.
4. The hotels are not currently using any of the suggested solutions.
5. The biggest factor affecting the effectiveness of earning income is booking cancellations.
6. Cancellations result in vacant rooms for the booked length of time.
7. Clients make hotel reservations the same year they make cancellations.

Research Question

1. What are the variables that affect hotel reservation cancellations?
2. How can we curb cancellations?
3. How will hotels be assisted in making pricing and promotional decisions?

Hypothesis

1. More cancellations occur when prices are higher.
2. When there is a longer waiting list, customers tend to cancel more frequently.
3. Longer lead time leads to higher cancellation rates
4. The majority of clients is coming from offline travel agents to make their reservations.
5. People are cancelling because they do not get the requested room, asked while booking.

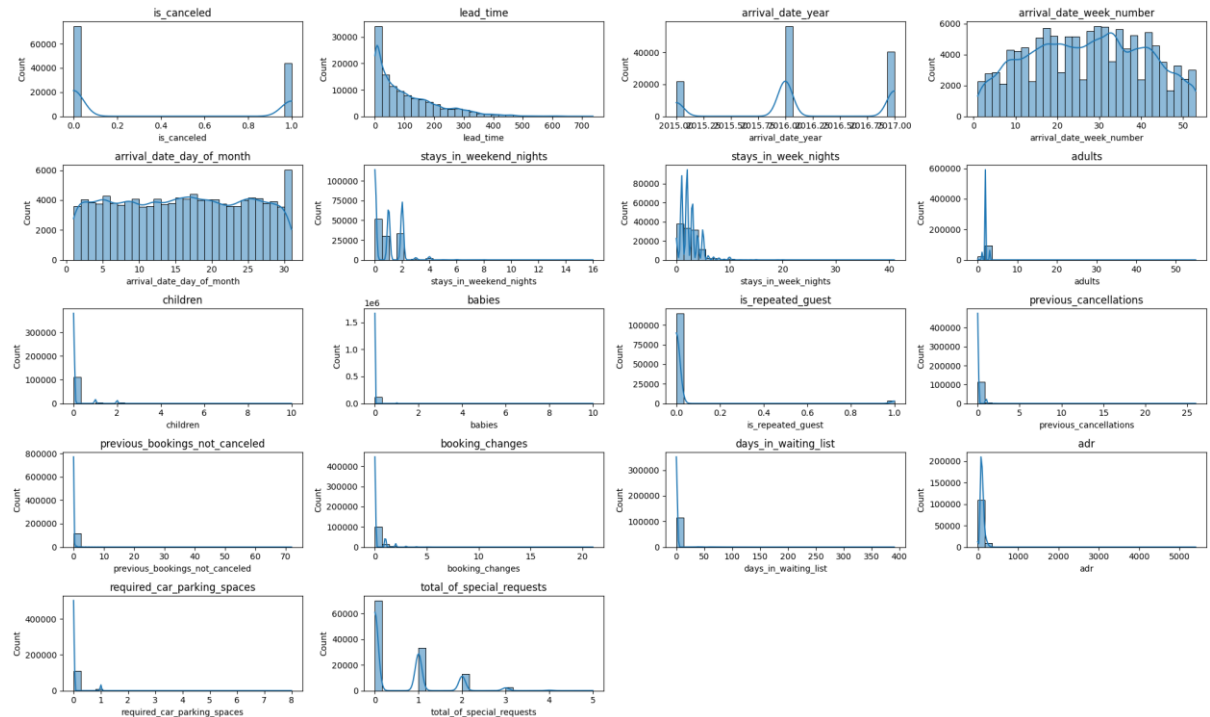
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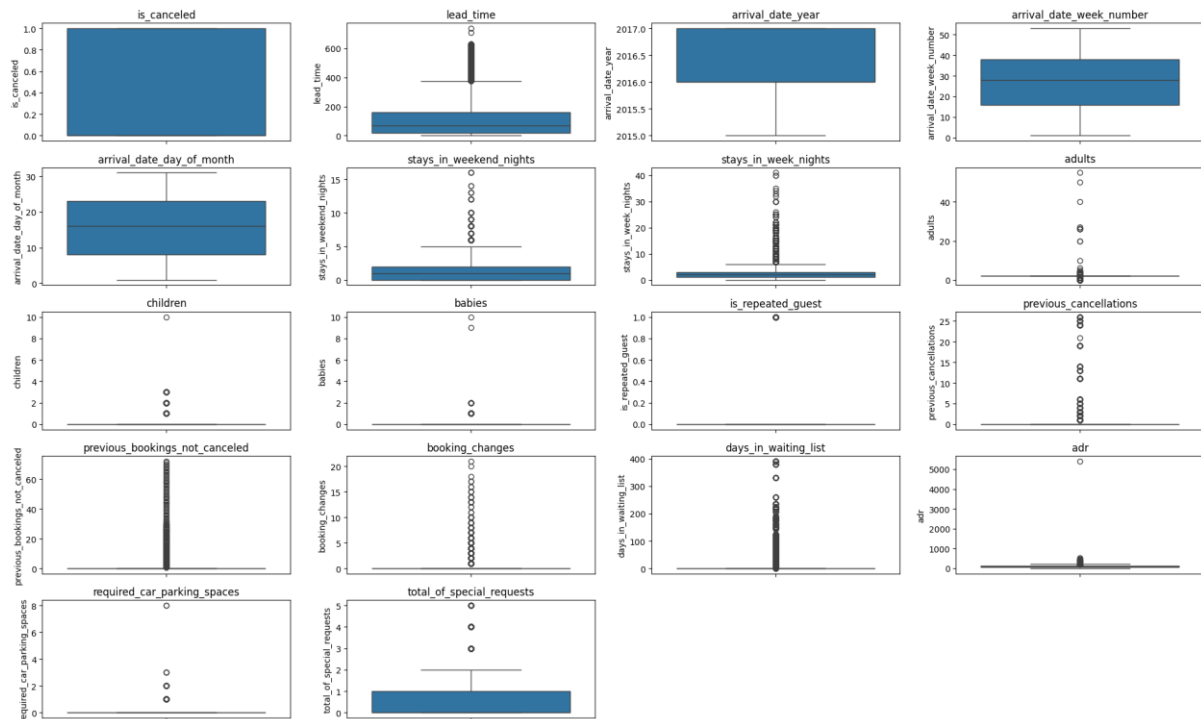
Summary Statistics

	hotel	arrival_date_month	meal	country	market_segment	distribution_channel	reserved_room_type	assigned_room_type	deposit_type	customer_type	reservation_status
count	118898	118898	118898	118898	118898	118898	118898	118898	118898	118898	118898
unique	2	12	5	177	7	5	10	12	3	4	3
top	City Hotel	August	BB	PRT	Online TA	TA/TO	A	A	No Deposit	Transient	Check-Out
freq	79302	13852	91863	48586	56402	97730	85601	73863	104163	89174	74745

	count	mean	min	25%	50%	75%	max	std
is_canceled	115608.0	0.363011	0.0	0.0	0.0	1.0	1.0	0.48087
lead_time	115608.0	94.9392	0.0	17.0	66.0	151.0	365.0	91.873736
arrival_date_year	115608.0	2016.151469	2015.0	2016.0	2016.0	2017.0	2017.0	0.71019
arrival_date_week_number	115608.0	27.059572	1.0	16.0	27.0	38.0	53.0	13.626287
arrival_date_day_of_month	115608.0	15.785854	1.0	8.0	16.0	23.0	31.0	8.779657
stays_in_weekend_nights	115608.0	0.938923	0.0	0.0	1.0	2.0	16.0	0.997596
stays_in_week_nights	115608.0	2.515172	0.0	1.0	2.0	3.0	41.0	1.913261
adults	115608.0	1.857069	0.0	2.0	2.0	2.0	55.0	0.584427
children	115608.0	0.107095	0.0	0.0	0.0	0.0	10.0	0.404301
babies	115608.0	0.008157	0.0	0.0	0.0	0.0	10.0	0.098661
is_repeated_guest	115608.0	0.032922	0.0	0.0	0.0	0.0	1.0	0.178432
previous_cancellations	115608.0	0.087771	0.0	0.0	0.0	0.0	26.0	0.856803
previous_bookings_not_canceled	115608.0	0.135293	0.0	0.0	0.0	0.0	72.0	1.505414
booking_changes	115608.0	0.223246	0.0	0.0	0.0	0.0	21.0	0.657063
days_in_waiting_list	115608.0	1.763018	0.0	0.0	0.0	0.0	178.0	12.137781
adr	115608.0	102.623221	-6.38	70.0	95.0	126.9	510.0	48.428864
required_car_parking_spaces	115608.0	0.06349	0.0	0.0	0.0	0.0	8.0	0.247086
total_of_special_requests	115608.0	0.581508	0.0	0.0	0.0	1.0	5.0	0.797637
reservation_status_date	115608	2016-07-31 02:08:13.253062144	2014-10-17 00:00:00	2016-02-07 00:00:00	2016-08-08 00:00:00	2017-02-08 00:00:00	2017-09-14 00:00:00	NaN
month	115608.0	6.29677	1.0	3.0	6.0	9.0	12.0	3.353342

Outlier Detection

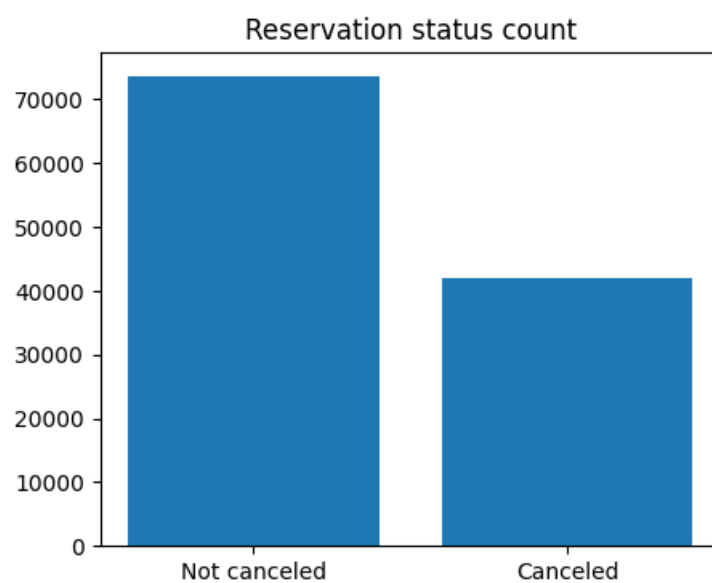




Outlier Treatment

1. Average Daily Rate < 5000
2. Lead Time < 365 days (1 yr)
3. Waiting period < 180 days (6 months)

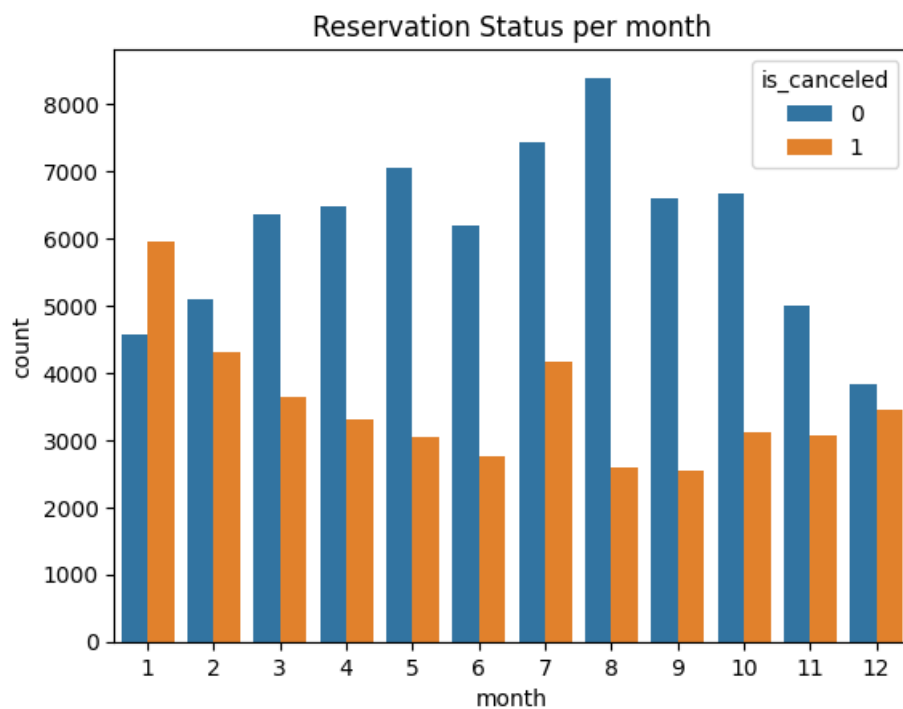
Analysis and Findings



The accompanying bar graph shows the percentage of reservations that are cancelled vs not cancelled. There are 37% of clients who cancelled their reservation, which has a significant impact on the hotels' earnings.

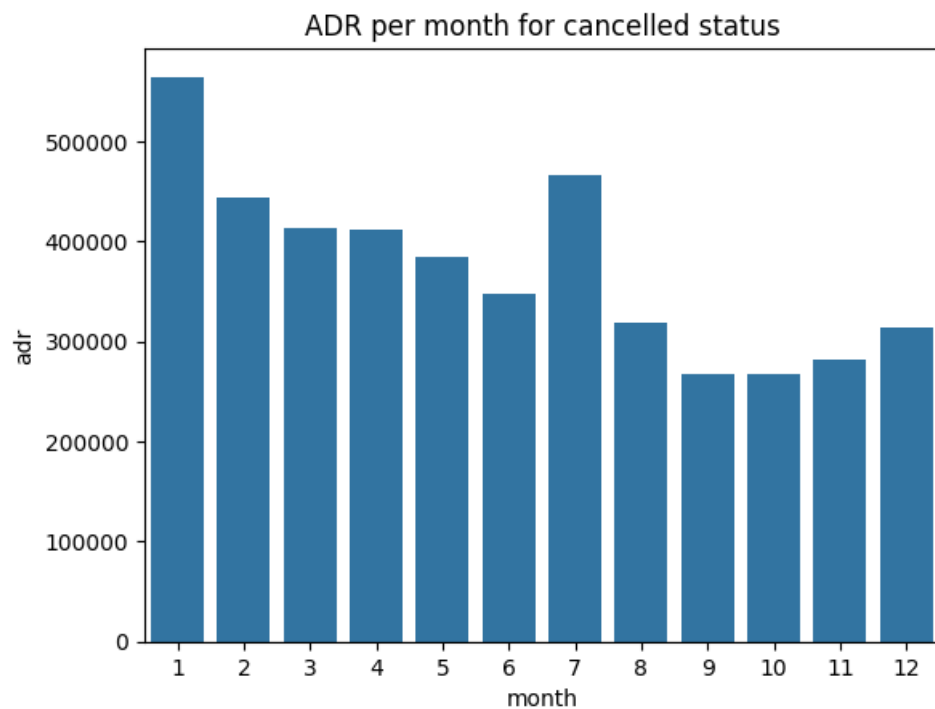


In comparison to resort hotels, city hotels have more bookings and higher cancellation rate of 40%. It's possible that city hotels have higher customer frequency.

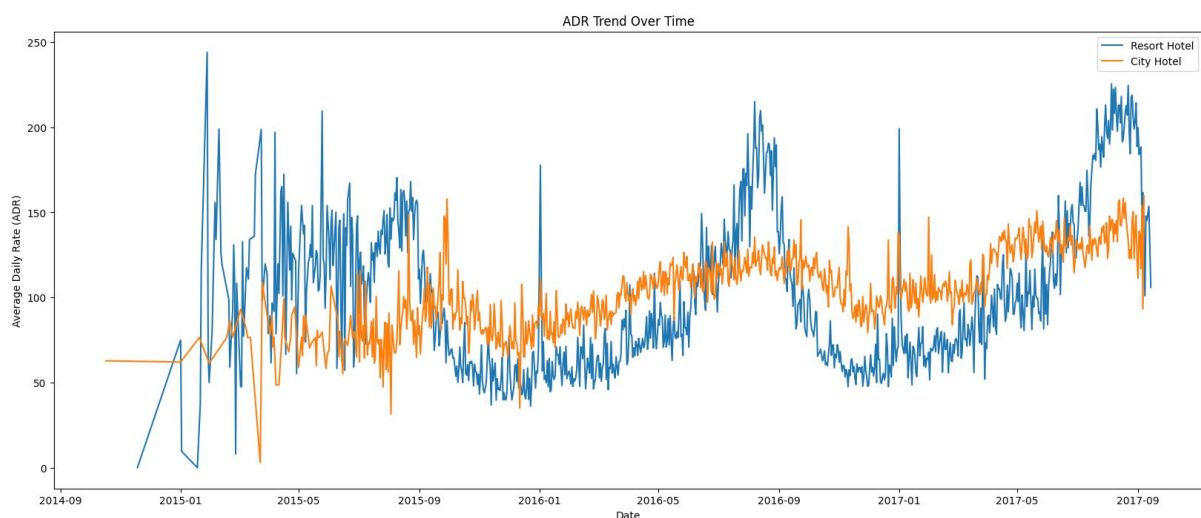


The above grouped bar analyses the months with the highest and lowest reservation levels according to reservation status. As can be seen, the number of confirmed reservations are largest in the month of August, whereas January is the month with the most cancelled reservations.

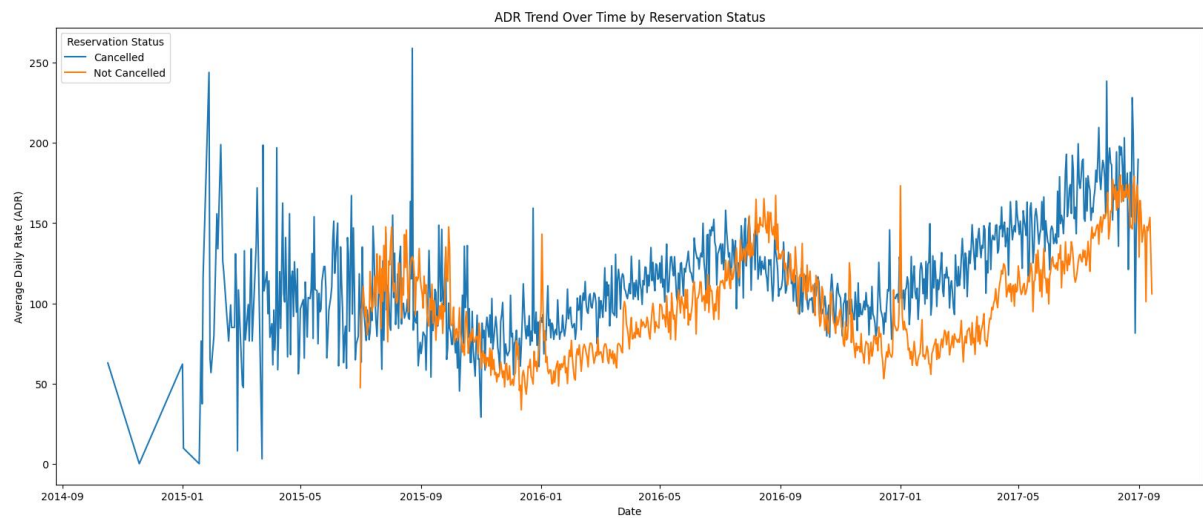
Average Daily Rate (ADR) Analysis



This bar graph demonstrates that cancellations are most common when prices are greatest and are least common when they are lowest. Therefore, the cost of the accommodation is mainly responsible for the cancellation. This intuition is further strengthened by the fact that city hotel has higher ADR and hence higher cancellation.



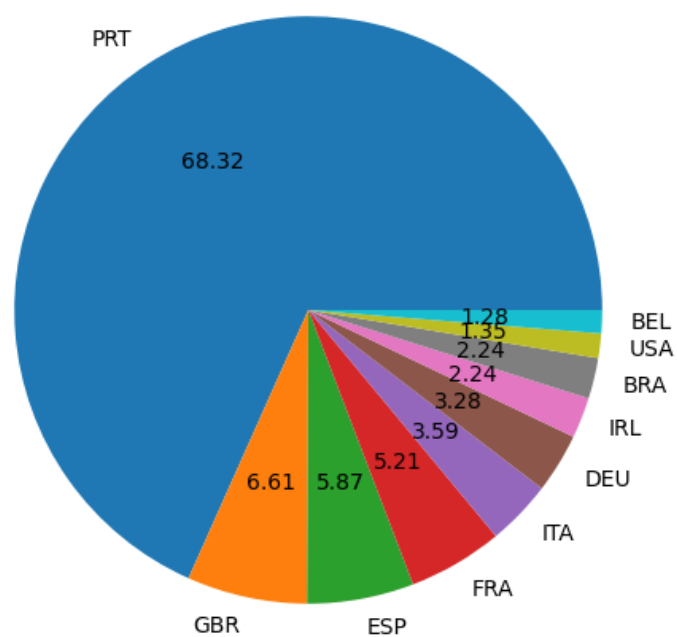
In general, the city hotel (106) has a higher average daily rate than resort hotel (96). However, there are some exceptions as shown in the above line chart.



As seen in the graph, cancelled reservations have higher ADR than the non-cancelled ones. It clearly proves all the above analysis, that the higher price leads to higher cancellation.

Country wise performance

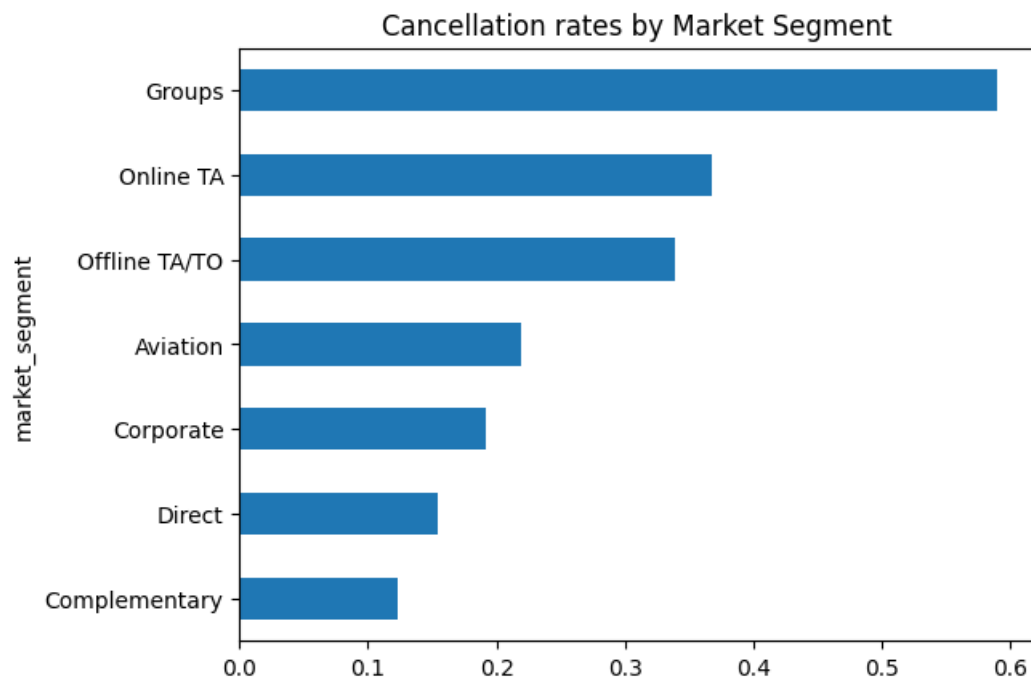
Top 10 countries with



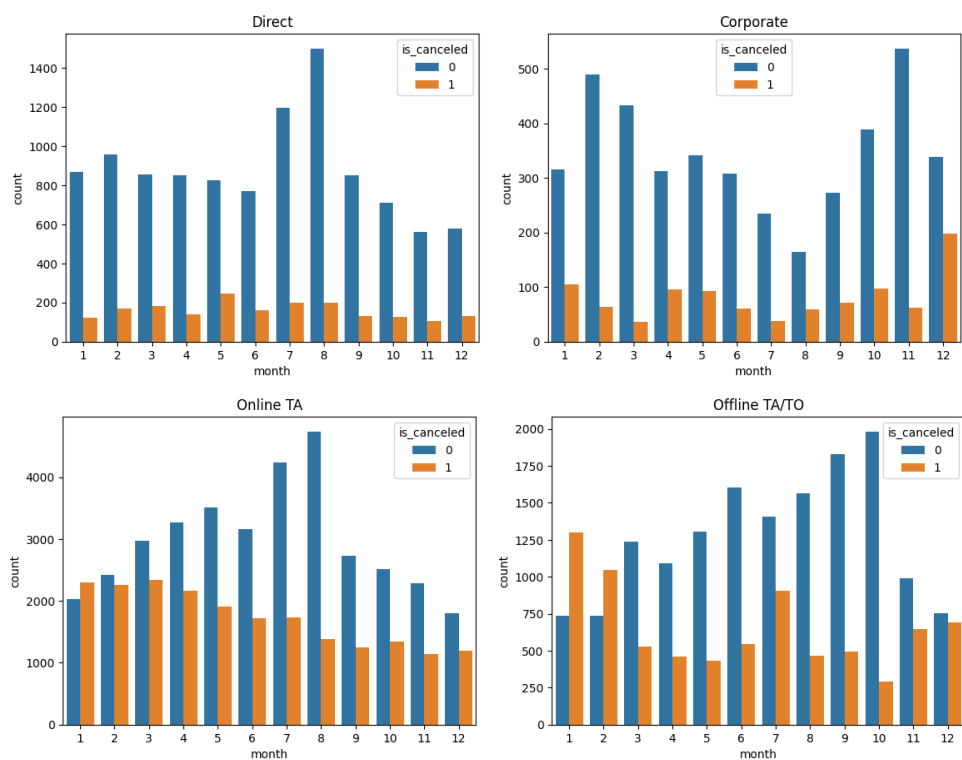
Portugal leads the chart with nearly 68% cancellation rates.

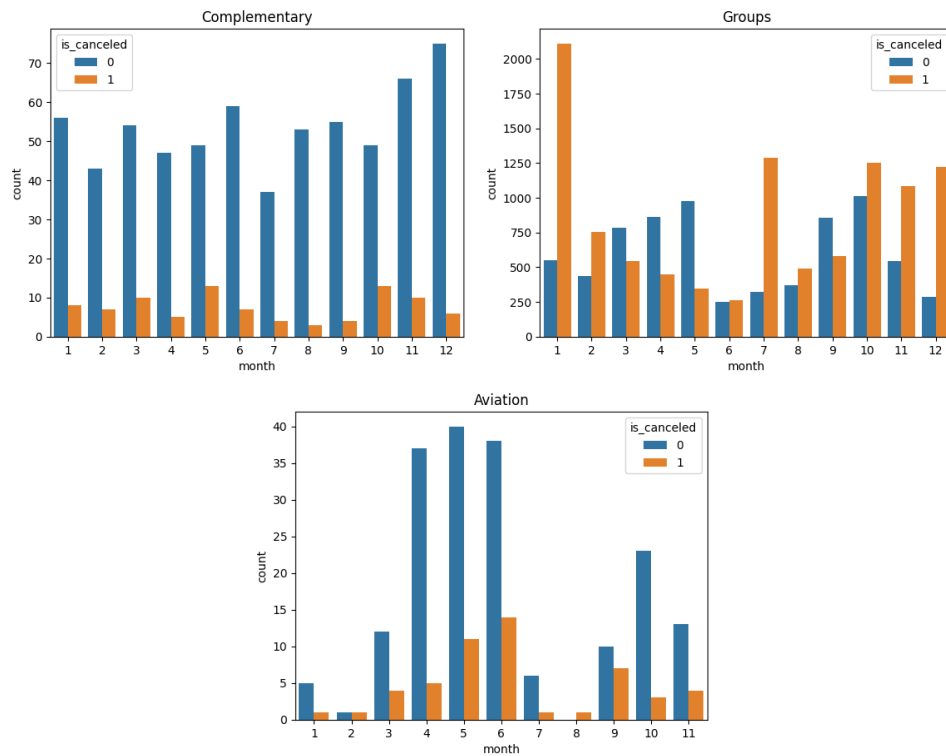
Market Segment Performance

Around 49% of the clients come from online travel agencies, whereas 15% come from groups. Only 10% of clients book hotels directly by visiting them and making reservations.



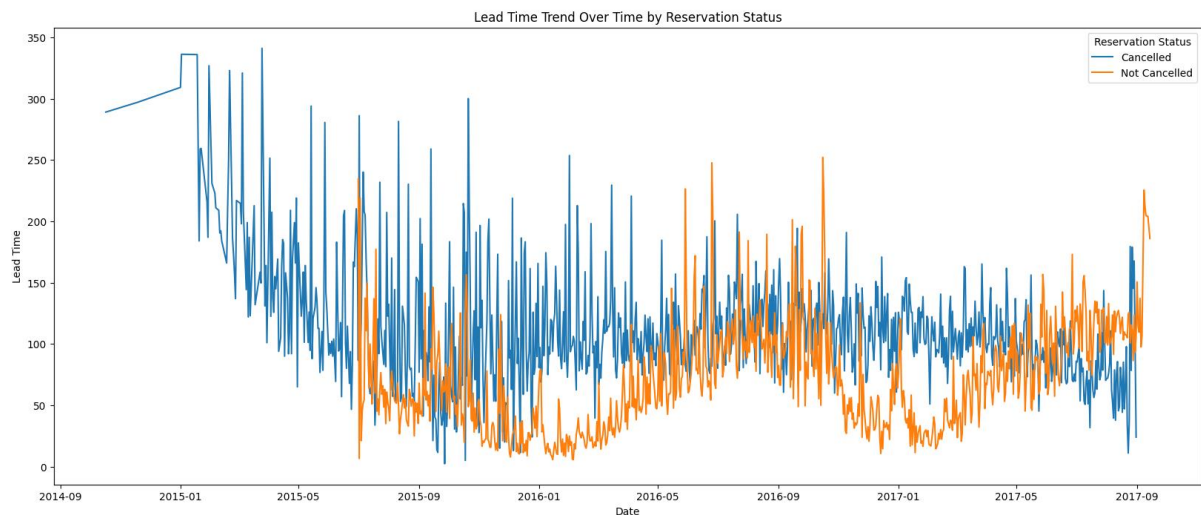
An interesting thing to note is that the people booking through groups have an abnormally high cancel rate.



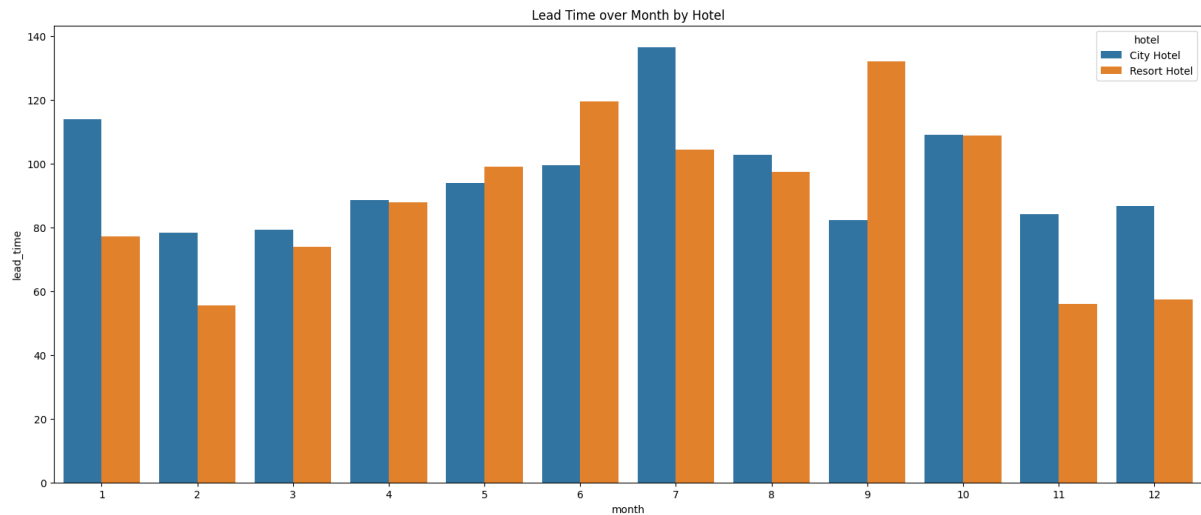


On further breakdown, it is clear that it is the Groups segment responsible for the high cancellations. Abnormal high numbers in the months of November to February, and July.

Lead Time Analysis



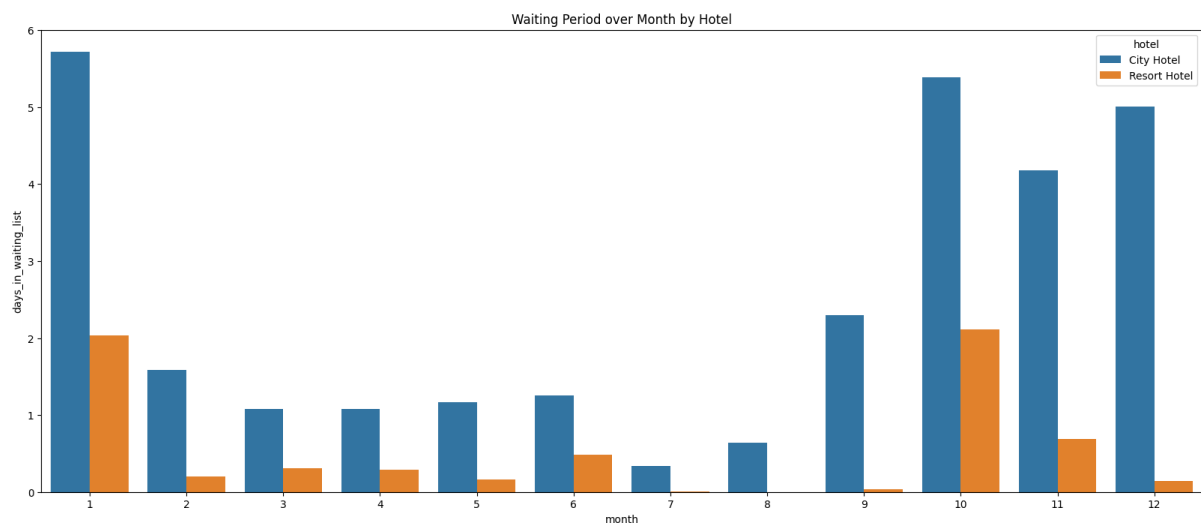
People cancelling have a higher lead time of 129 days compared to 75 days of non-cancelling clients. Suggesting clients shift to better deals, tougher market competition or change in travel.



This analysis is further confirmed by comparing the lead time of city and resort hotel. City hotel has consistently high lead time, especially during November to February, months when cancellations are at peak .

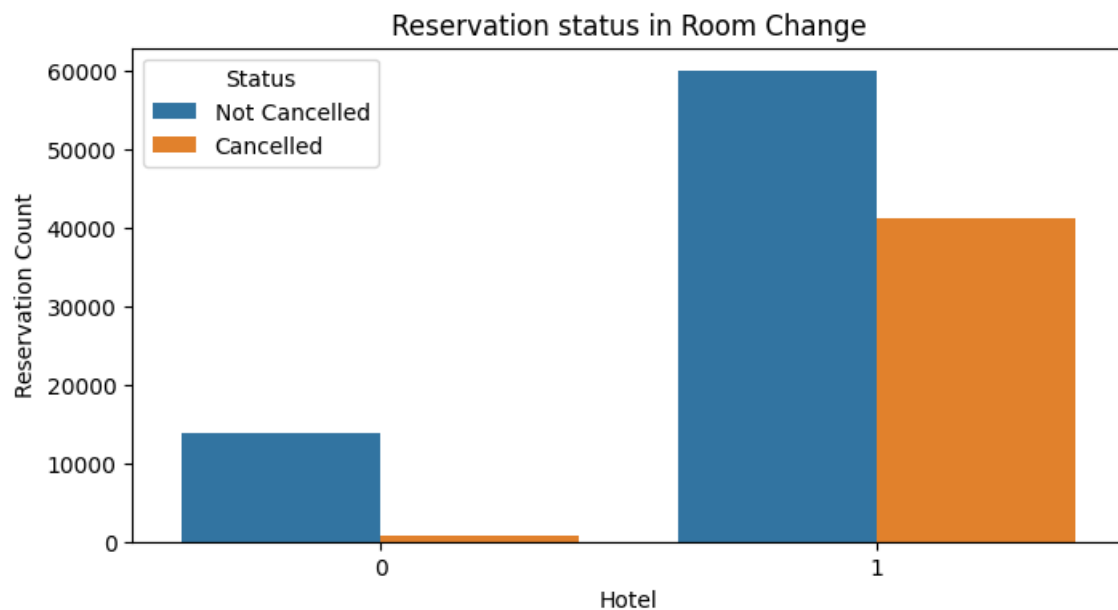
Waiting Period Analysis

Average waiting period for cancelling clients is 2.8 days compared to 1.16 days of non-cancelling clients. Indicating swifter operation and confirmation time is good for customer acquisition, and the market is tough.



Waiting time is very important, as it is evident from the above graph. City hotel has a waiting period of 2.4 days and resort hotels only 0.5 days. City hotel also has a higher customer frequency, as suggested earlier, this implies that city hotels has some operational inefficiencies which is causing such high waiting period.

Room Miss-Match



12% of the customers do not get the rooms they ask for. Interestingly, only 5% of these people cancel their booking, this number is higher for people with requested room. This suggests people are not frustrated enough to cancel their bookings if they do not receive their requested rooms.

Suggestions

1. Cancellation rates rise as the price does. In order to prevent cancellations of reservations, hotels could work on their pricing strategies and try to lower the rates for specific hotels based on locations. They can also provide some discounts to the consumers.
2. Device a strategy to regulate groups/bulk bookings, because they are the largest population contributing to cancellation numbers.
3. Higher lead time leads to loss of customers and affects room availability. So lead times should be lowered by restricting people from booking rooms in too advance, which prevents unnecessary demand and keeps ADR low.
4. People need quick confirmation on their bookings. Streamline booking acknowledgement process through tech inclusion in operations to utilize the maximum potential of available rooms.